CIRCUIT main board — schematic status

Drawn (draft, from KiCad library pinouts): 2 USB-C input; 3 Haptics, buttons, debug

Not drawn yet:
- Charger, battery, fuel gauge; 3V3 buck-boost, load switches, backlight driver;
  secure element; NFC; BLE module — these need pinouts from datasheets (no library symbols)
- Main MCU and display/touch connectors — blocked on gate G1 and panel selection

Signals between sheets use global labels. No routing before this schematic is complete and reviewed.

USB-C input, CC, ESD

File: usb.kicad_sch

Haptics, buttons, debug

File: hmi.kicad_sch

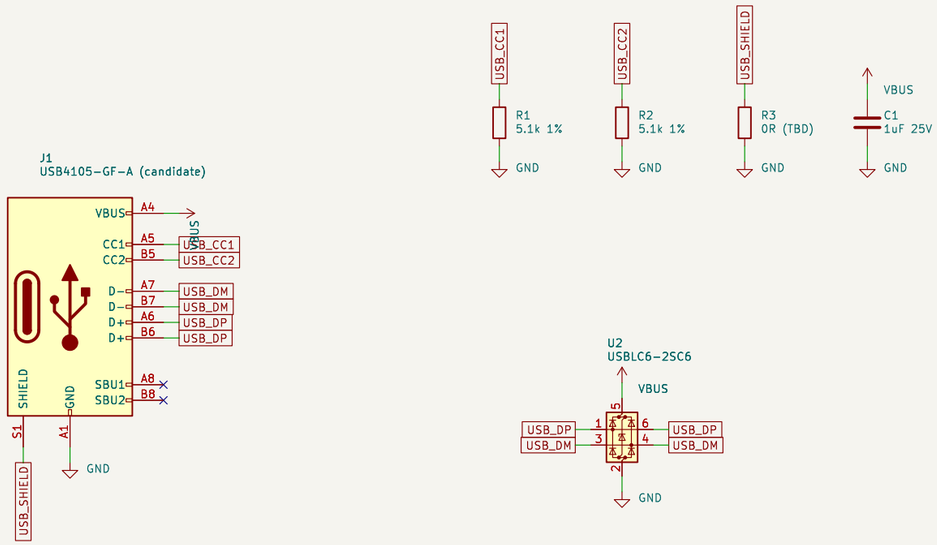
DRAFT — generated by hardware/scripts/gen_schematic.py — not reviewed

Sheet: /
File: circuit.kicad_sch

Title: Circuit main board

Size: A3	Date:	**Rev: A0-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/3

USB-C INPUT (5 V sink, USB 2.0 device, no Power Delivery)
- R1/R2: 5.1k Rd on each CC pin separately (USB Type-C sink requirement).
- USB_CC1/USB_CC2 go to MCU ADC inputs to read source current (0.5/1.5/3 A).
- U2 clamps D+/D- and VBUS. Both pins of each I/O pair are one node: route through the part.
- C1 kept small: a sink may present at most 10 uF on VBUS.
TODO before review: 20 V-tolerant CC protection (TPD4S311-class, no library symbol yet);
VBUS over-voltage is handled by the charger input rating - confirm from charger datasheet;
shield-to-ground scheme (R3) to be decided with the enclosure ESD path.



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Sheet: /USB-C input, CC, ESD/
File: usb.kicad_sch

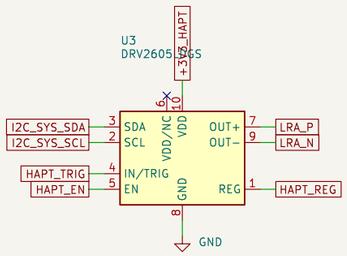
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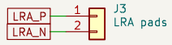
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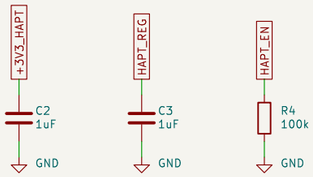
HAPTICS: DRV2605L drives an 8 mm LRA from the switched rail +3V3_HAPT.
R4 holds EN low while the MCU is in reset. Pin 6 (VDD/NC) left open — CHECK datasheet.
C2/C3 values (1 uF) are defaults — CHECK datasheet.

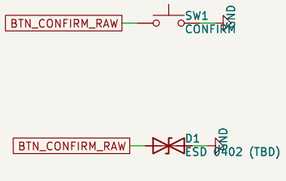
BUTTONS: active low. Pull-up, series resistor, RC debounce and ESD clamp per button.
The MCU reads BTN_CONFIRM_N directly in the secure world; it is the only signing approval input.
BTN_POWER_RAW also goes to the charger push-button input (power sheet, not drawn yet).
Switch and TVS part numbers TBD (candidates: Panasonic EVQ-P7 side-push; 0402 ESD diode).

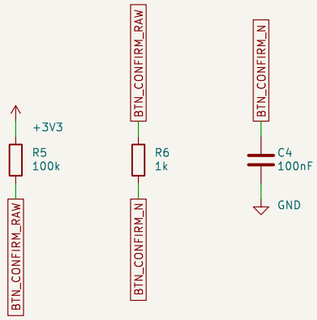
DEBUG: Tag-Connect TC2030-NL pads,
no connector fitted. Internal only.
Production: debug locked in the MCU;
pads not reachable with the case closed.

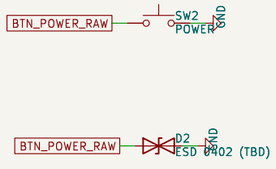


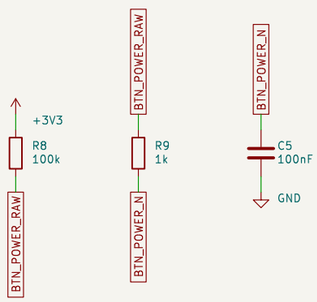


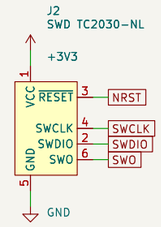












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